Cyber-Physical Agent Orchestrator

Architecture concept | In progress

Company application: I contribute to a server assembly application at my company.

Concept

Coordinate AI planning, knowledge retrieval, visual feedback and edge equipment execution. Observations inform the next plan in a closed feedback loop.

Agent responsibilities

Planner: Coordinate tasks and revise plans.
KM: Retrieve task specifications, CAD geometry and prior knowledge.
Physical: Execute permitted edge actions and report equipment limits.
VLM: Interpret images and return findings or optical feedback.

Example application

Server assembly inspection: reflection or shadow can prompt a revised lighting or capture plan before another observation.

Status

Architecture concept in progress. Broader use across objects and equipment remains a development direction. Implementation, equipment limits and performance require validation.

Repository: github.com/smes112108/cyber-physical-agent-orchestrator